

In-Class Exercise/Demo



In-Class Demo

- In this demo, we will practice [React-Router](#).
- First, create a new repository [m5-w2-d2-demo](#) in your remote Github and then clone this in your WESTCLIFF folder.
- We will [first scaffold a React app](#). Launch VS Code and navigate to the clone folder.
- In a bash terminal, run [npx create-react-app demo-app](#)
- Next, head over to GAP and download [m5-w2-d2-demo.zip](#) folder, unzip and drop the content into the demo-app - *src* folder.
- Return to the bash terminal and cd into demo-app: [cd demo-app](#)



In-Class Demo #1

- With the react app now setup, our next step is to install the react-router.
- On the bash terminal, run `npm install react-router-dom`
- Open `routerexample1.js` in the src folder.
- Import `BrowserRouter Router, Switch, Route, Link`

```
import {  
  BrowserRouter as Router,  
  Switch,  
  Route,  
  Link  
} from "react-router-dom";
```



In-Class Demo

- Now let's write our export function to show our links.
- Inside `<Router>` on line 8 add `<div>`
- We are creating these links so we can later set Paths for it.

```
<div>
  <nav>
    <ul>
      <li>
        <Link to="/">Home</Link>
      </li>
      <li>
        <Link to="/about">About</Link>
      </li>
      <li>
        <Link to="/users">Users</Link>
      </li>
    </ul>
  </nav>
</div>
```



In-Class Demo

- Now let's write our switch statement to handle the routes.

```
<div>
  .....
  <Switch>
    <Route path="/about">
      <About />
    </Route>
    <Route path="/users">
      <Users />
    </Route>
    <Route path="/">
      <Home />
    </Route>
  </Switch>
</div>
```



In-Class Demo

Complete code

```
<div>
  <nav>
    <ul>
      <li>
        <Link to="/">Home</Link>
      </li>
      <li>
        <Link to="/about">About</Link>
      </li>
      <li>
        <Link to="/users">Users</Link>
      </li>
    </ul>
  </nav>
  <Switch>
    <Route path="/about">
      <About />
    </Route>
    <Route path="/users">
      <Users />
    </Route>
    <Route path="/">
      <Home />
    </Route>
  </Switch>
</div>
```



In-Class Demo

Next, We are going to write our function for each route.

To add multiple lines we need add (html here); to return. Like this example

```
function Home() {  
  return (  
    <div>  
      <h2>Home</h2>  
      <p>This is home</p>  
    </div>  
  );  
}  
  
function About() {  
  return <h2>About</h2>;  
}  
  
function Users() {  
  return <h2>Users</h2>;  
}
```



In-Class Demo

- Now, let's update our `index.js` file to import `routerexample1`
- On line 4, let's comment out `'./App'`
- Add import from `routerexample1`

```
//import App from './App';  
import App from './routerexample1'
```



In-Class Demo

- Run `npm start`
- It will open <http://localhost:3000/>
- Click through pages

- [Home](#)
- [About](#)
- [Users](#)

Home

This is home



In-Class Demo #2

- Open `routerexample2.js` file from the `src` folder.
- Let's import `BrowserRouter Router, Switch, Route, Link, useRouteMatch, useParams`

```
import {  
  BrowserRouter as Router,  
  Switch,  
  Route,  
  Link,  
  useRouteMatch,  
  useParams  
} from "react-router-dom";
```



In-Class Demo

- Now let's review our Router and Switch

```
<div>
  <ul>
    <li>
      <Link to="/">Home</Link>
    </li>
    <li>
      <Link to="/about">About</Link>
    </li>
    <li>
      <Link to="/topics">Topics</Link>
    </li>
  </ul>

  <Switch>
    <Route path="/about">
      <About />
    </Route>
    <Route path="/topics">
      <Topics />
    </Route>
    <Route path="/">
      <Home />
    </Route>
  </Switch>
</div>
```



In-Class Demo

Next, We are going to write our function for each route.

```
function Home() {  
  return <h2>Home</h2>;  
}  
  
function About() {  
  return <h2>About</h2>;  
}
```



In-Class Demo

For Topic, We are going to code a little different so we can have other links present on Topic page.

```
function Topics() {  
  let match = useRouteMatch();  
  
  return (  
    <div>  
      <h2>Topics</h2>  
  
      <ul>  
        <li>  
          <Link to={`${match.url}/cats`} >Cats</Link>  
        </li>  
        <li>  
          <Link to={`${match.url}/dogs`} >  
            Dogs  
          </Link>  
        </li>  
      </ul>  
    </div>  
  );  
}
```



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Let's add a `switch` statement to handle the paths on topic page.

```
</ul>

<Switch>
  <Route path={` ${match.path}/:topicId`} >
    <Topic />
  </Route>
  <Route path={match.path}>
    <h3>Please select a topic.</h3>
  </Route>
</Switch>
</div>
```



In-Class Demo

Complete Return code

```
return (  
  <div>  
    <h2>Topics</h2>  
  
    <ul>  
      <li>  
        <Link to={`${match.url}/cats`} >Cats</Link>  
      </li>  
      <li>  
        <Link to={`${match.url}/dogs`} >  
          Dogs  
        </Link>  
      </li>  
    </ul>  
  
    <Switch>  
      <Route path={`${match.path}/:topicId`} >  
        <Topic />  
      </Route>  
      <Route path={match.path}>  
        <h3>Please select a topic.</h3>  
      </Route>  
    </Switch>  
  </div>  
) ;
```



In-Class Demo

Next add a [parameter](#) with the topic id

```
function Topic() {  
  let { topicId } = useParams();  
  return <h3>Requested topic ID: {topicId}</h3>;  
}
```



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- Now, let's update our `index.js` file to import `routerexample2`
- On line 5, let's comment out `'./routerexample1'`
- Add import from `routerexample2`

```
//import App from './App';  
//import App from './routerexample1'  
import App from './routerexample2'
```



In-Class Demo

- Refresh the page on <http://localhost:3000/>
- Click through pages

- [Home](#)
- [About](#)
- [Topics](#)

Home

Select Topics

- [Home](#)
- [About](#)
- [Topics](#)

Topics

- [Cats](#)
- [Dogs](#)

Please select a topic.

Select a Topic
You will see the URL changing.
<http://localhost:3000/topics/cats>

- [Home](#)
- [About](#)
- [Topics](#)

Topics

- [Cats](#)
- [Dogs](#)

Requested topic ID: cats



In-Class Exercise Demo

Due:

Today 10.30PM PT

Submission:

Post Github link on GAP Week 2 Day 2 Exercise